



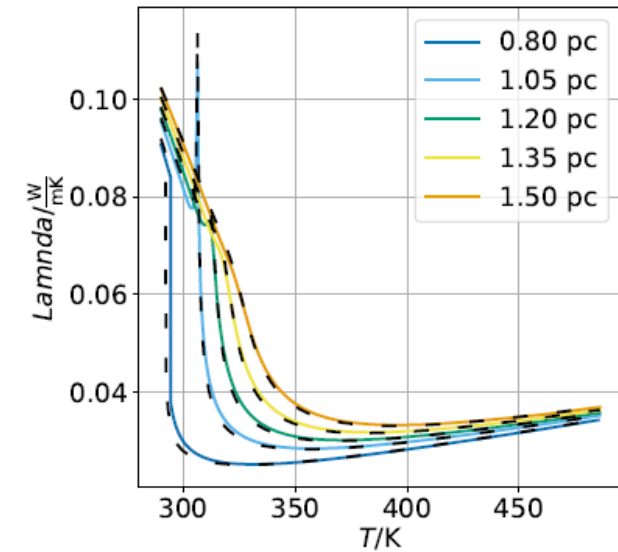
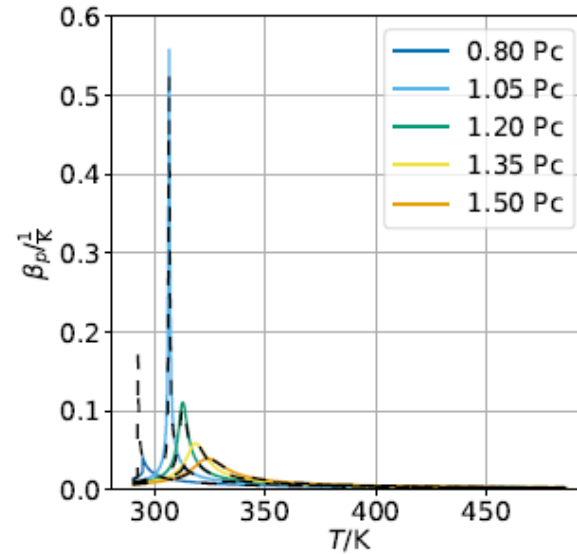
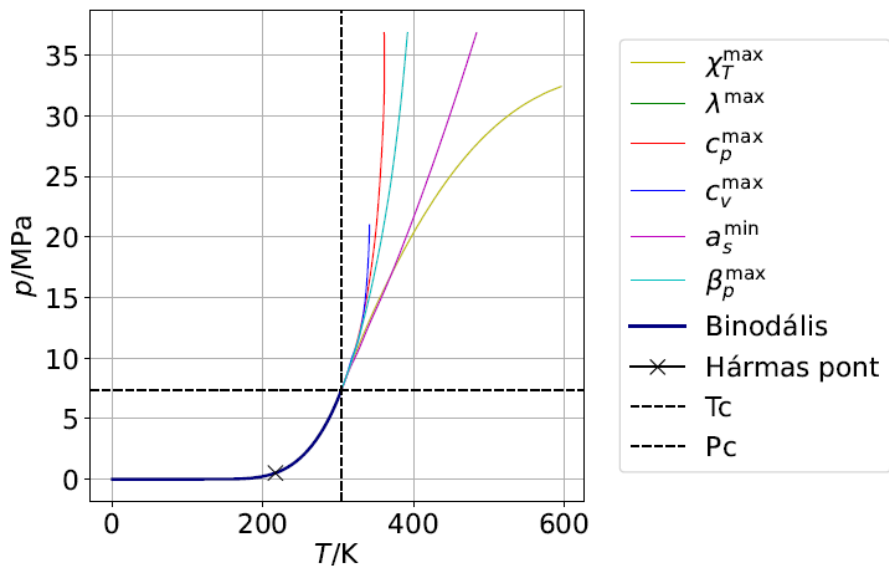
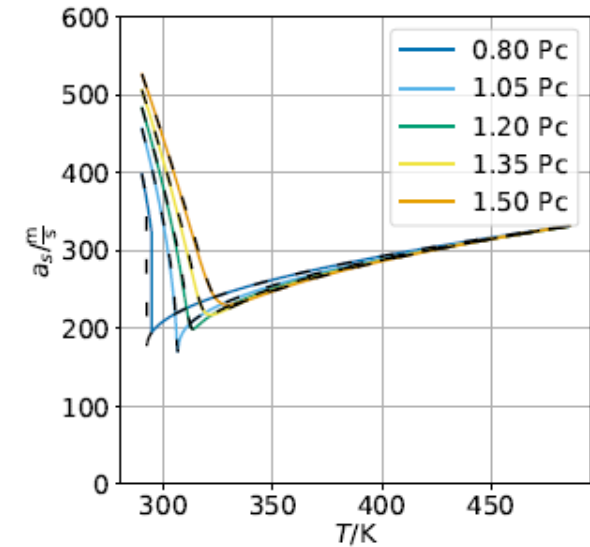
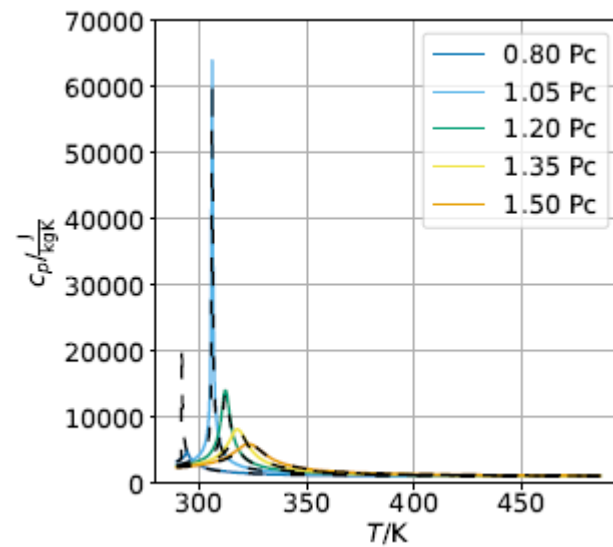
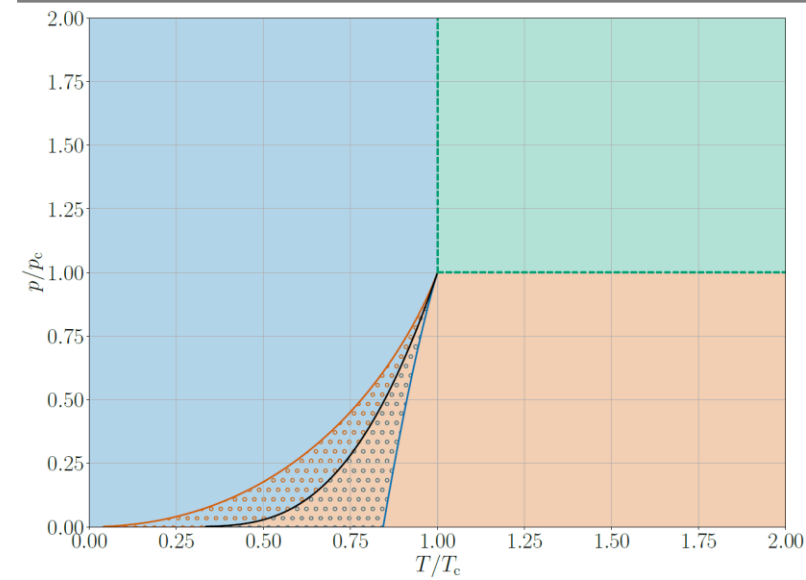
A szuperkritikus dugattyúhatás kezdeti feltételektől való függésének vizsgálata

Tóth Kristóf Ferenc

Gépészmérnök BSc 7. félév

Témavezető: Szücs Mátyás

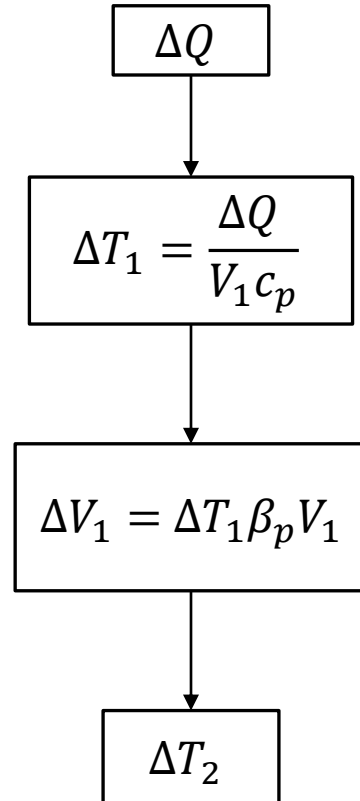
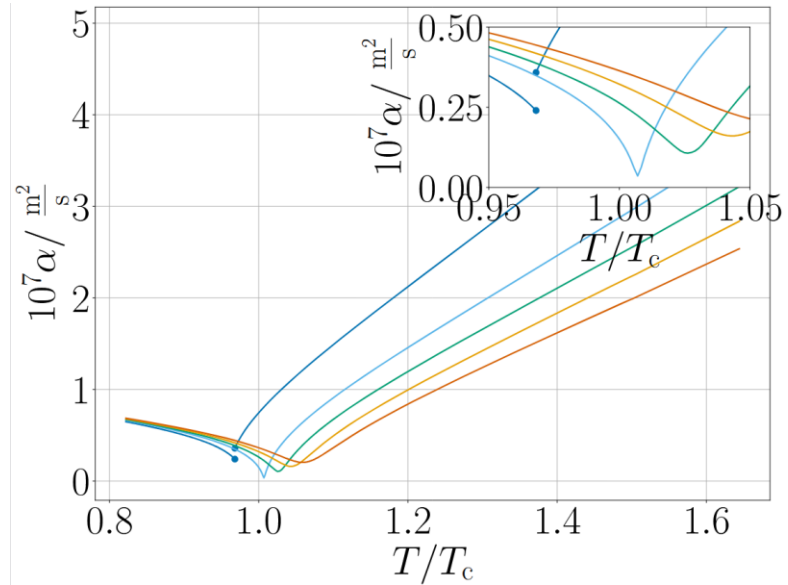
Szuperkritikus fluidum



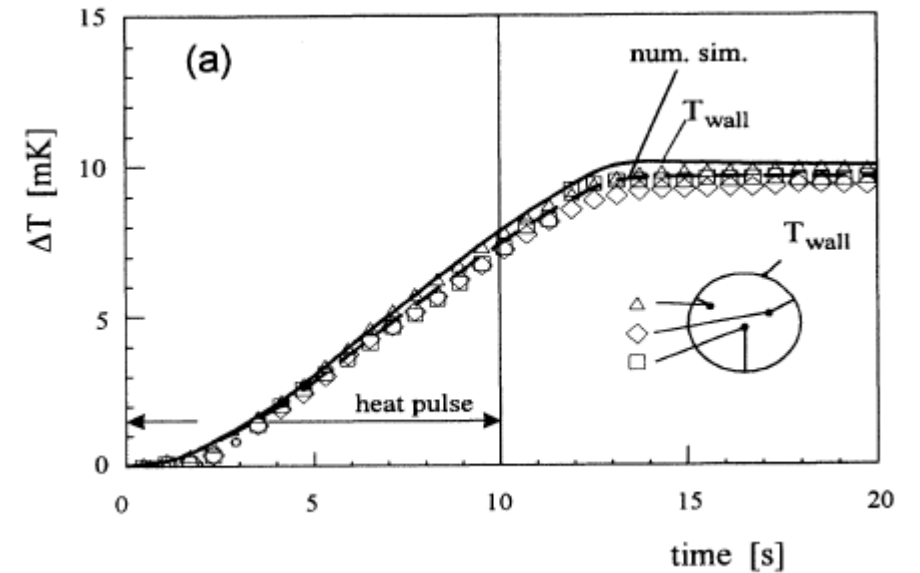
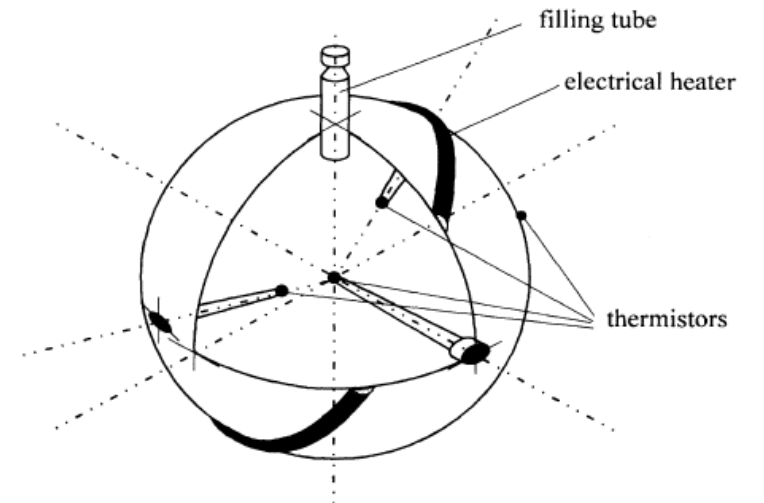
Dugattyúhatás

$$\tau_{\text{diff}} = \frac{\ell_{\text{char}}^2}{\lambda / \rho c_p}$$

Onuki 1990:



Straub 1995:



A dugattyúhatás matematikai modellje

Boukari 1990:

$$\rho c_p \frac{\partial T}{\partial t} = -\frac{\partial \dot{q}}{\partial x} + T \beta_p \frac{dp}{dt},$$

$$\dot{q} = -\lambda \frac{\partial T}{\partial x},$$

$$\frac{dp}{dt} = \frac{\int_0^L \rho \beta_p \frac{\partial T}{\partial t} dx}{\int_0^L \rho \chi_T dx}$$

Peremfeltételek:

$$\dot{q}(t, x = 0) = \begin{cases} \dot{q}_0, & \text{ha } 0 \leq t \leq t_h, \\ 0, & \text{ha } t_h \leq t, \end{cases} \quad \dot{q}(t, x = L) = 0$$

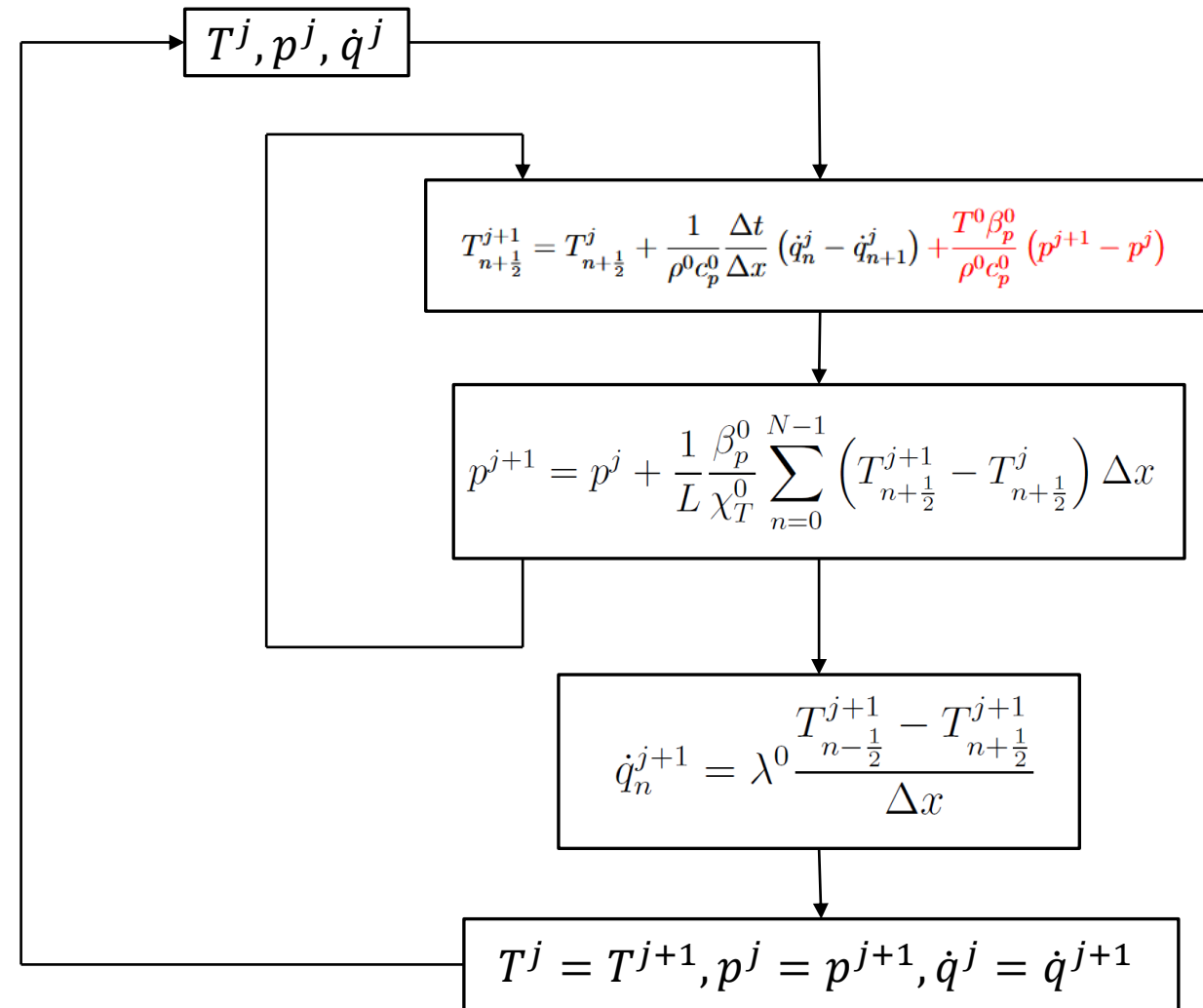
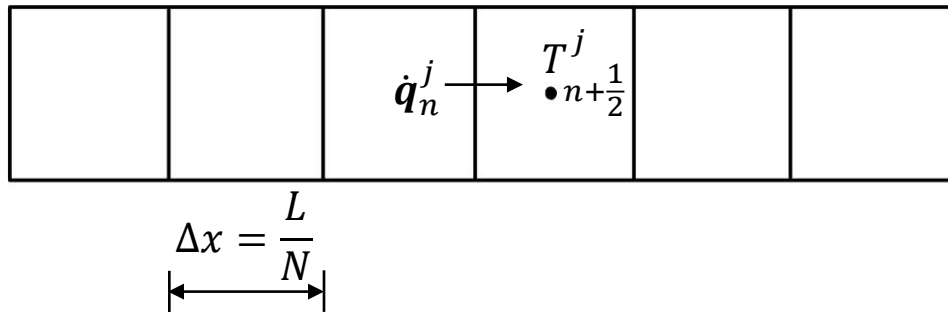
Kezdeti feltételek: T^0, p^0

Lineáris eset vizsgálata

$$\rho^0 c_p^0 \frac{\partial T}{\partial t} = -\frac{\partial \dot{q}}{\partial x} + T^0 \beta_p^0 \frac{dp}{dt},$$

$$\frac{dp}{dt} = \frac{1}{L} \frac{\beta_p^0}{\chi_T^0} \int_0^L \frac{\partial T}{\partial t} dx,$$

$$\dot{q} = -\lambda^0 \frac{\partial T}{\partial x}$$



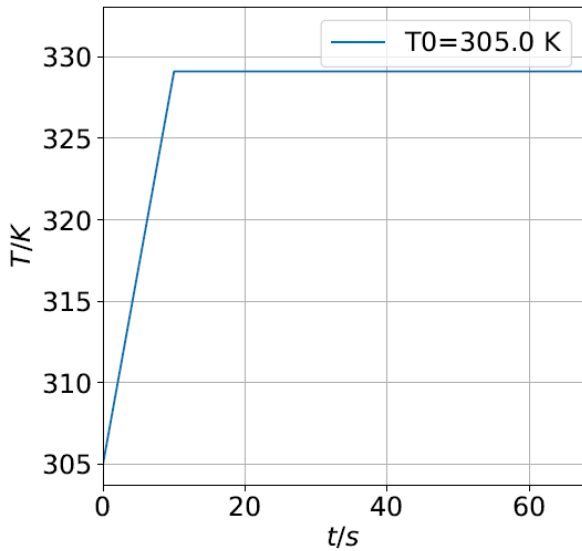
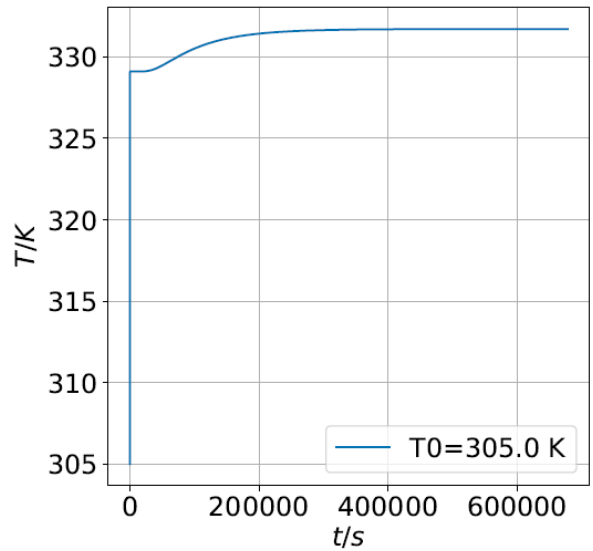
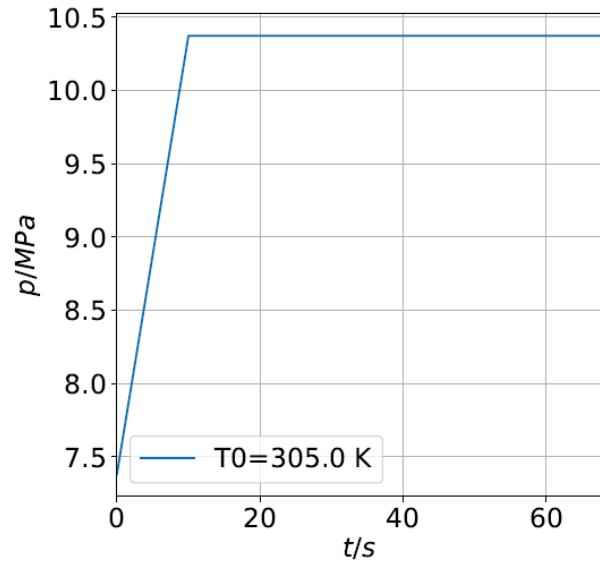
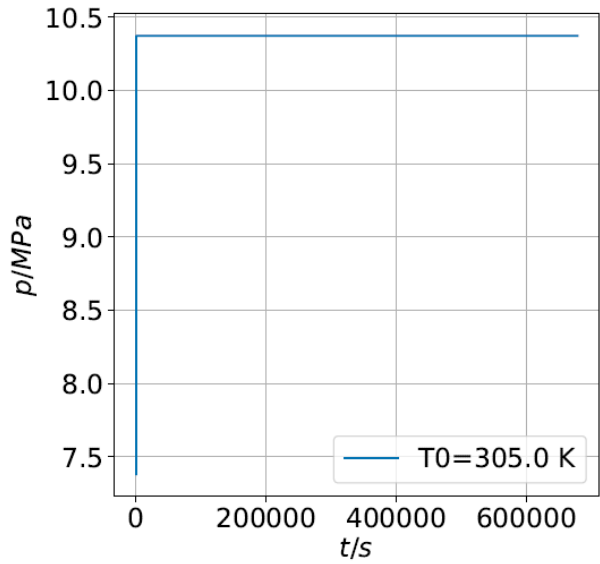
Lineáris eset vizsgálata

$$T^0 = 305 \text{ K}$$

$$p^0 = 7.38 \text{ MPa}$$

$$Q = 38.5 \text{ J}$$

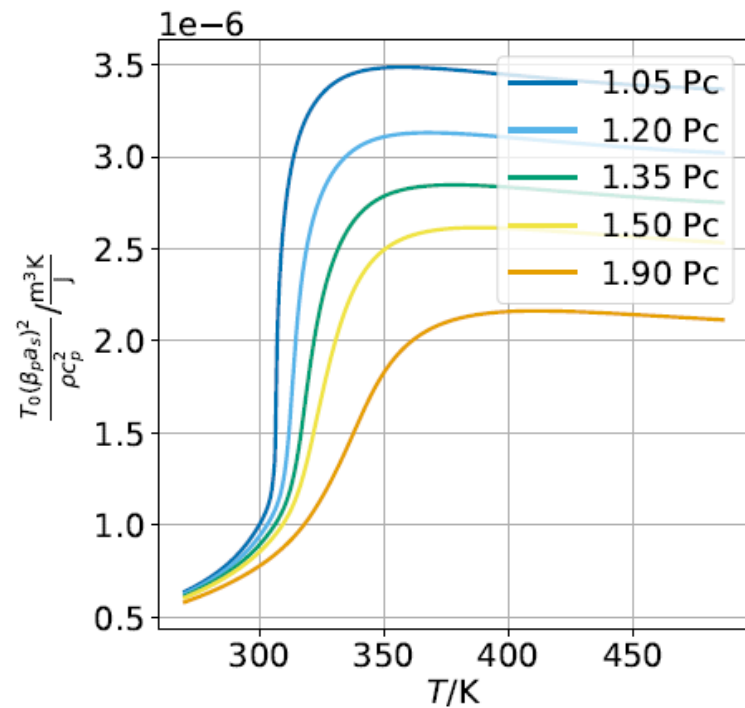
$$t_h = 10 \text{ s}$$



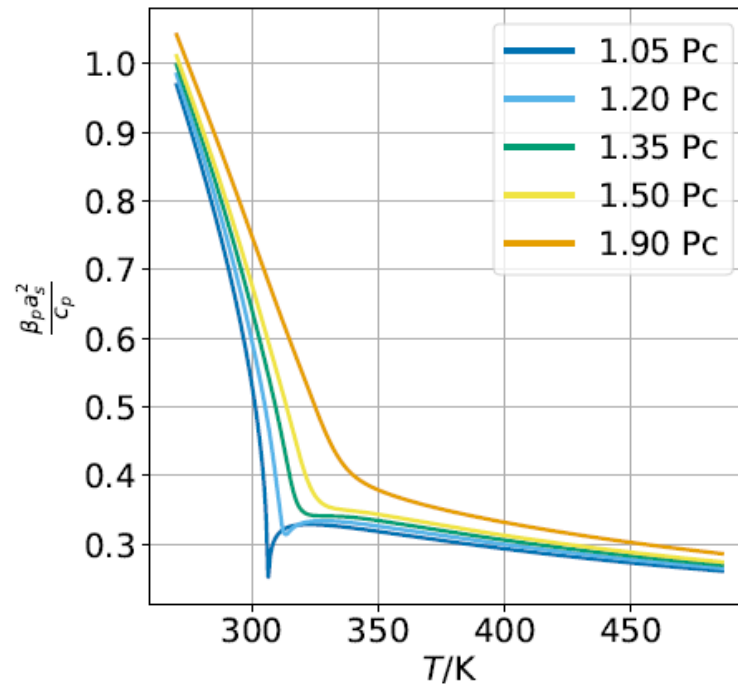
Domináns tagok a lineáris egyenletekben

$$T^{0+} - T^0 = \frac{Q}{\frac{4\pi}{3}R_g^3} \frac{T^0}{\rho^0} \left(\frac{\beta_p^0 a_s^0}{c_p^0} \right)^2, \quad p^\infty - p^0 = \frac{\beta_p^0}{\chi_T^0} (T^\infty - T^0) = \frac{Q}{\frac{4\pi}{3}R_g^3} \frac{\beta_p^0}{\chi_T^0 \rho^0 c_v^0}, \quad T^\infty - T^0 = \frac{Q}{\frac{4\pi}{3}R_g^3} \frac{1}{\rho^0 c_v^0}$$

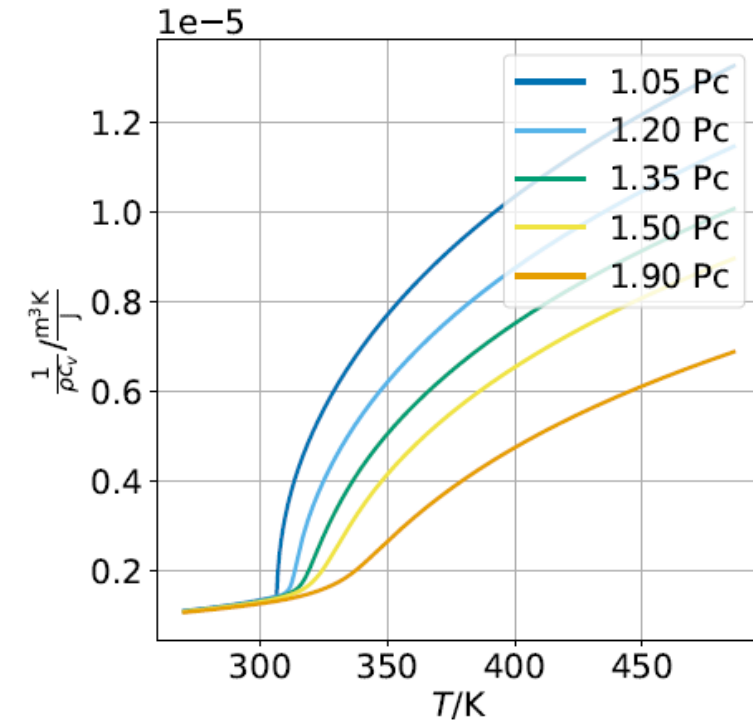
$$\frac{T^0}{\rho^0} \left(\frac{\beta_p^0 a_s^0}{c_p^0} \right)^2$$



$$\frac{\beta_p^0 (a_s^0)^2}{c_p^0}$$



$$\frac{1}{\rho^0 c_v^0}$$

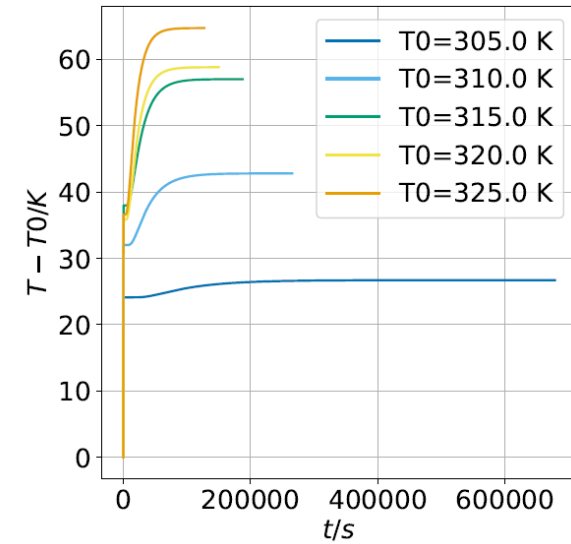
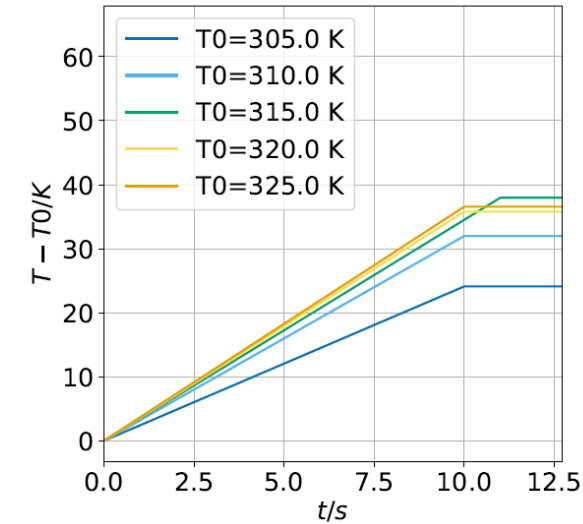
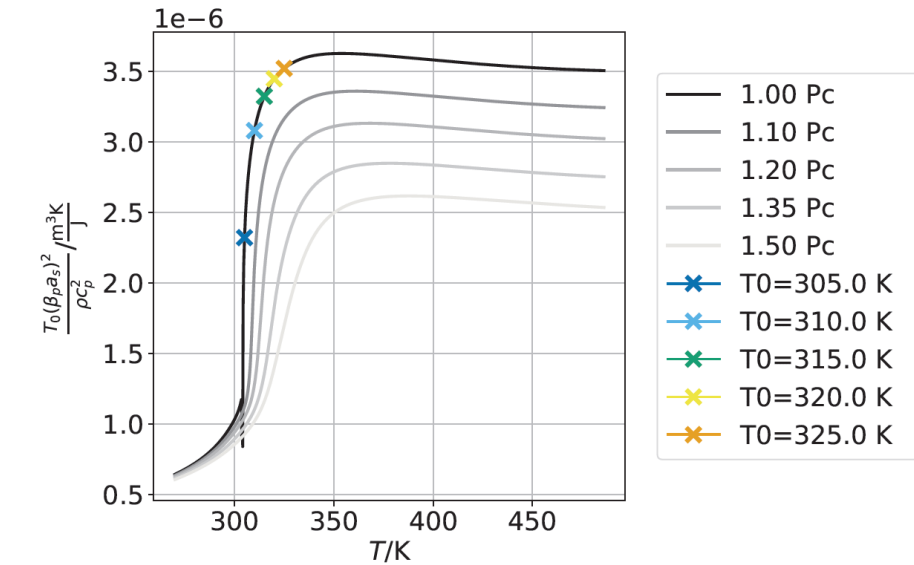
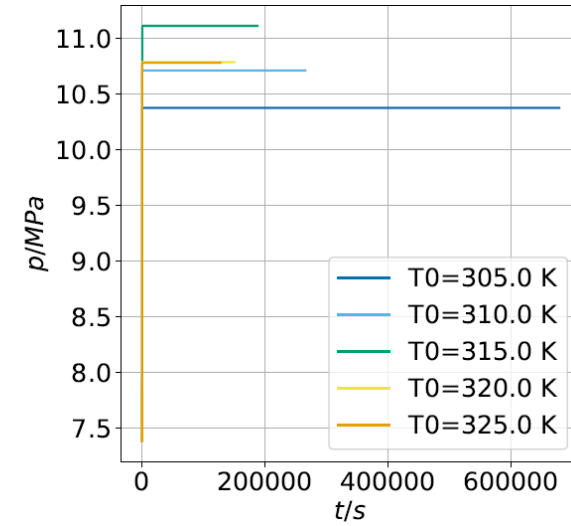
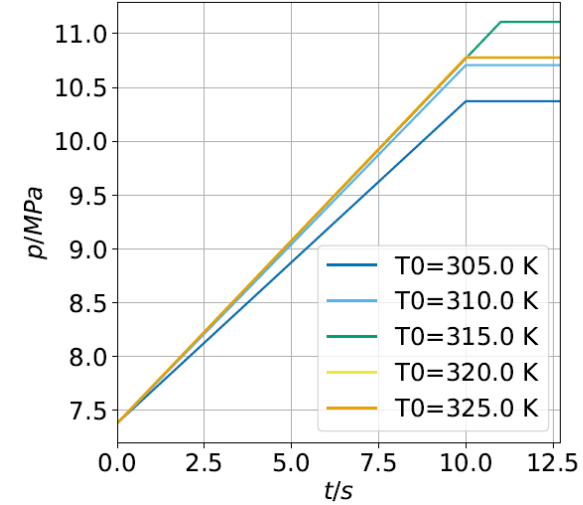
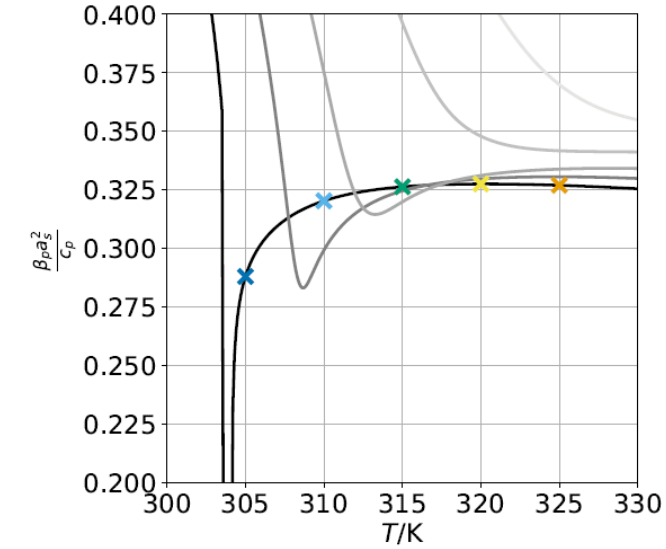


Lineáris eset vizsgálata

Q/J	t/s	$\frac{T^{0+}-T^0}{\text{K}}$	$\frac{T^{0+}-T^0}{\text{K}}$	$\frac{T^\infty-T^0}{\text{K}}$	$\frac{T^\infty-T^0}{\text{K}}$	$\frac{p^\infty-p^0}{\text{Pa}}$	$\frac{p^\infty-p^0}{\text{Pa}}$
		számított	szimulált	számított	szimulált	számított	szimulált
0,0385	10	0,0241	0,0241	0,0267	0,0267	2992,0	2991,4
38,5	10	24,1044	24,1044	26,7145	26,7142	2992041	2992040
38,5	2	24,1044	24,1044	26,7145	26,7142	2992041	2992040

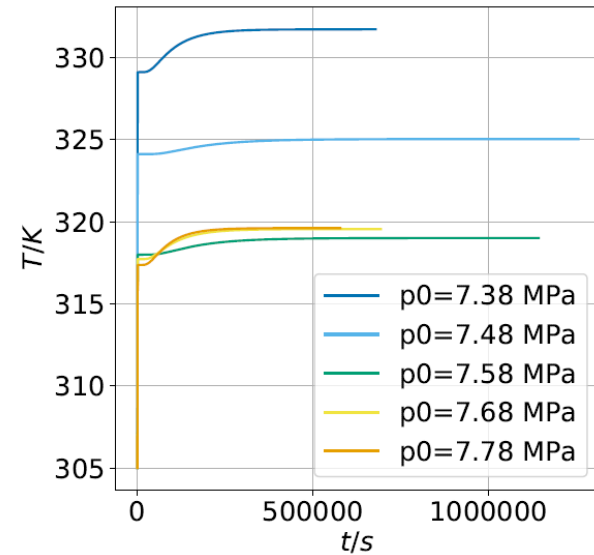
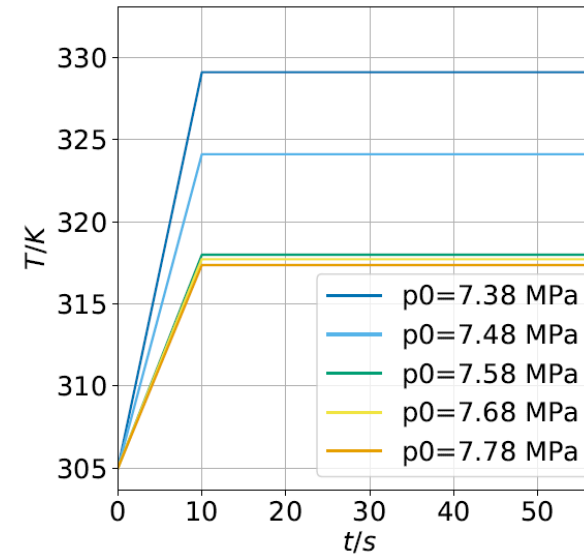
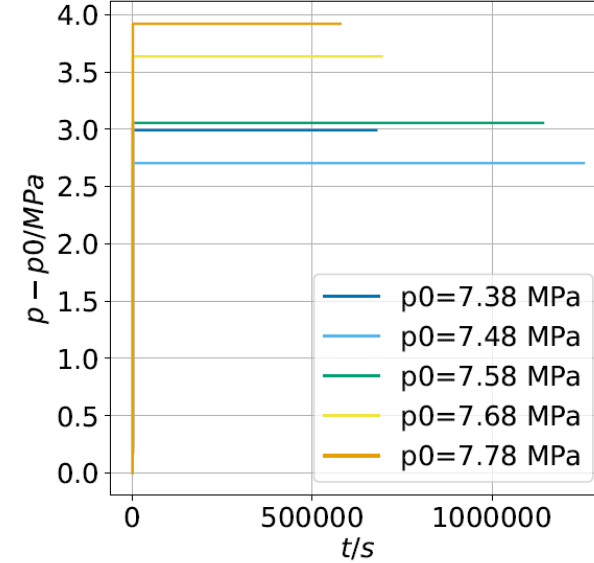
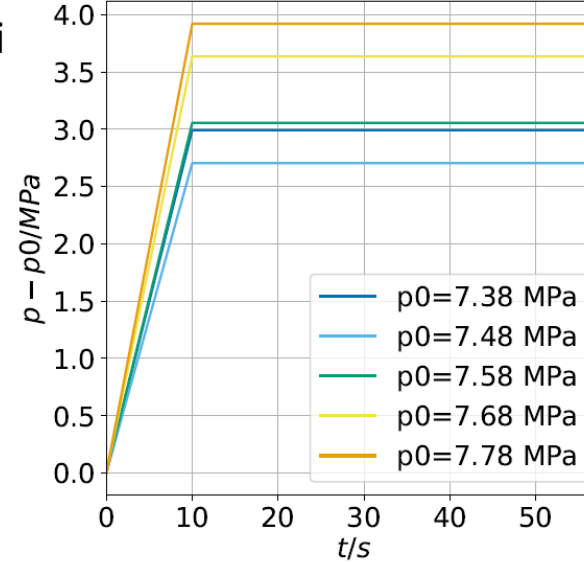
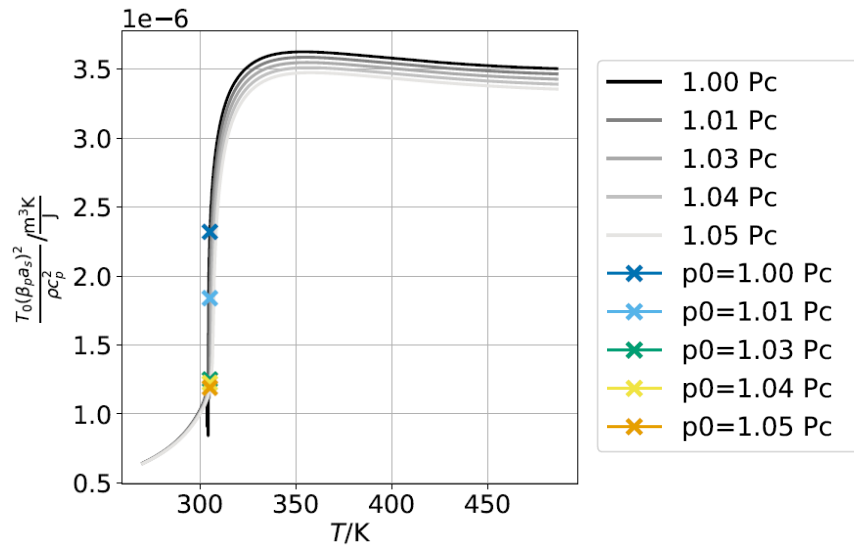
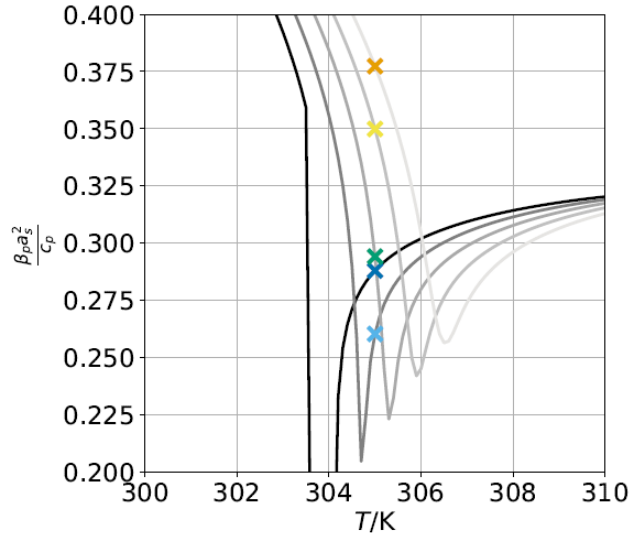
Kezdeti feltételek hatásának vizsgálata

- Azonos, $p^0 = 7.38$ MPa kezdeti nyomású esetek



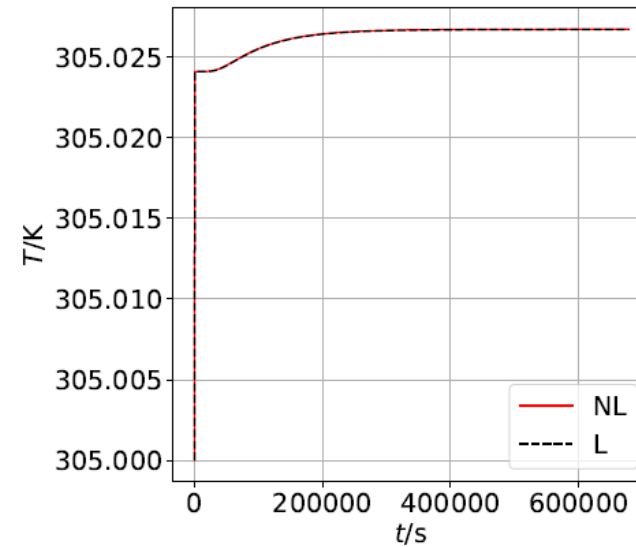
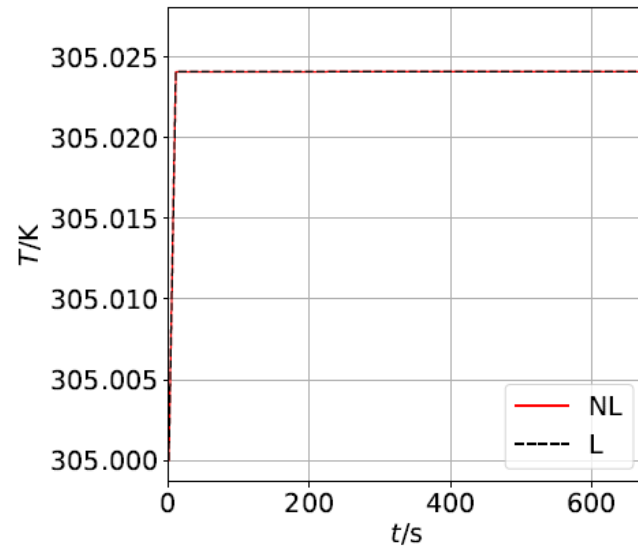
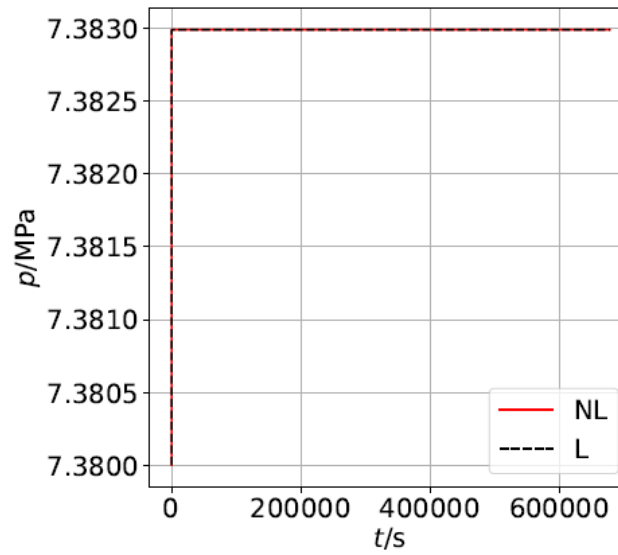
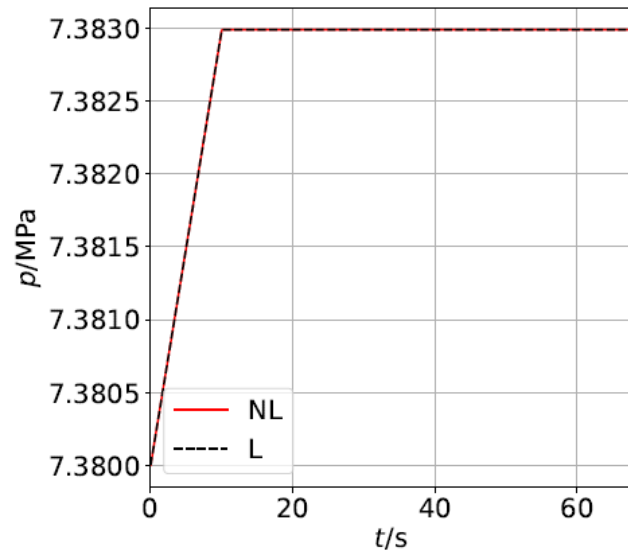
Kezdeti feltételek hatásának vizsgálata

- Azonos, $T^0 = 305$ K kezdeti hőmérsékletű esetek

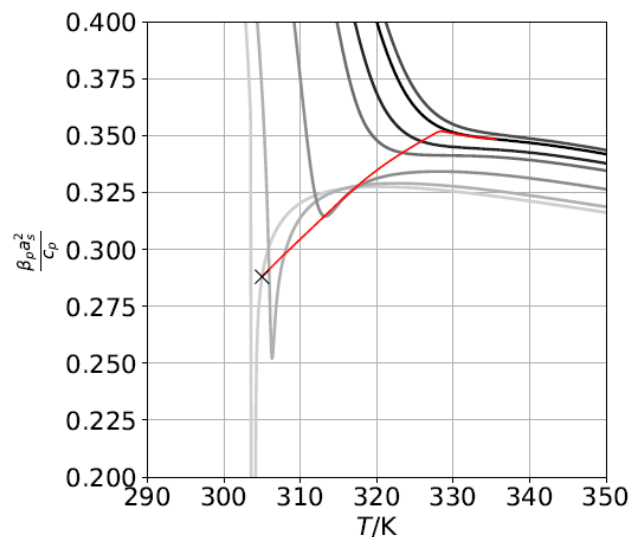


Nemlineáris eset vizsgálata

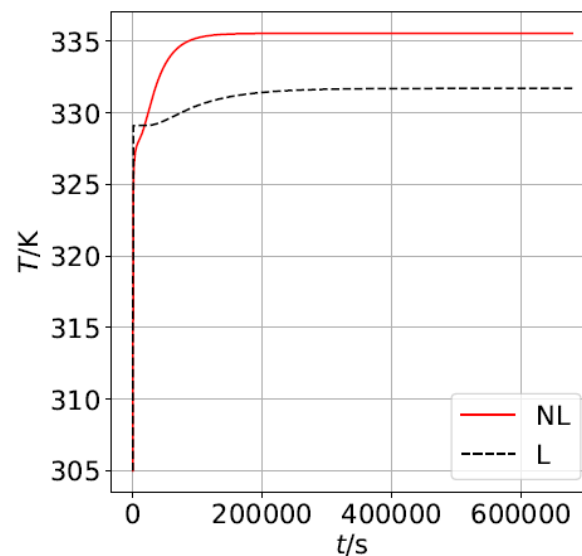
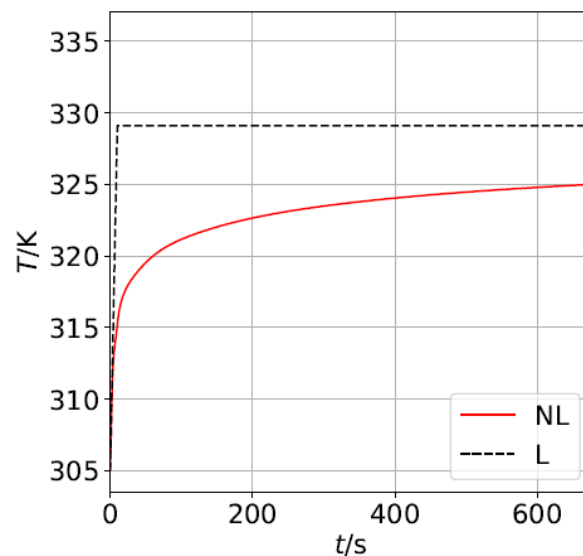
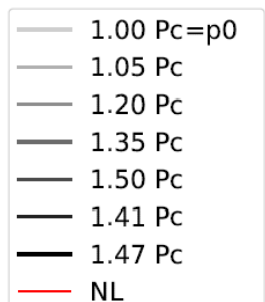
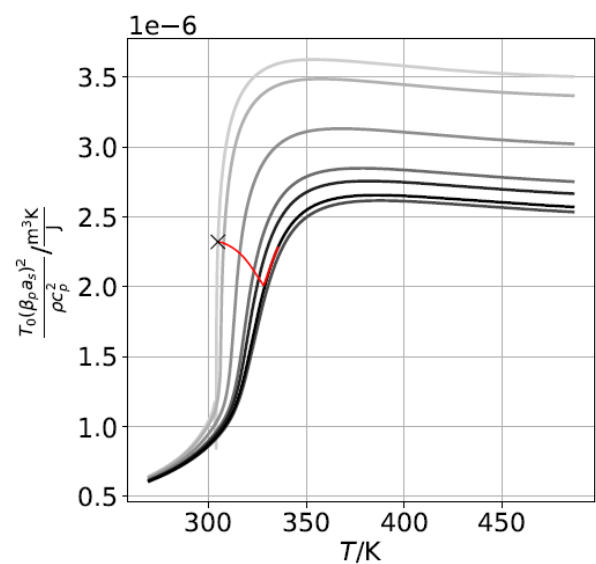
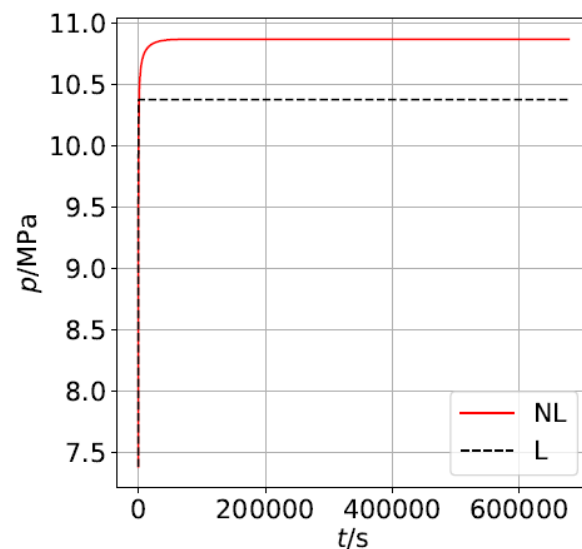
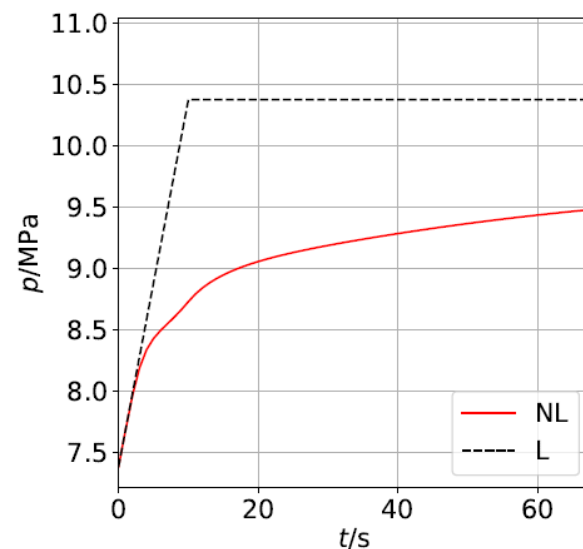
$T^0 = 305 \text{ K}$
 $p^0 = 7.38 \text{ MPa}$
 $Q = 0.0385 \text{ J}$
 $t_h = 10 \text{ s}$



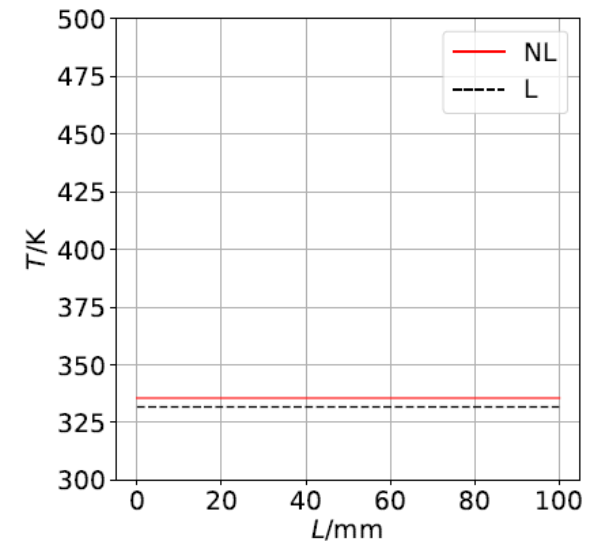
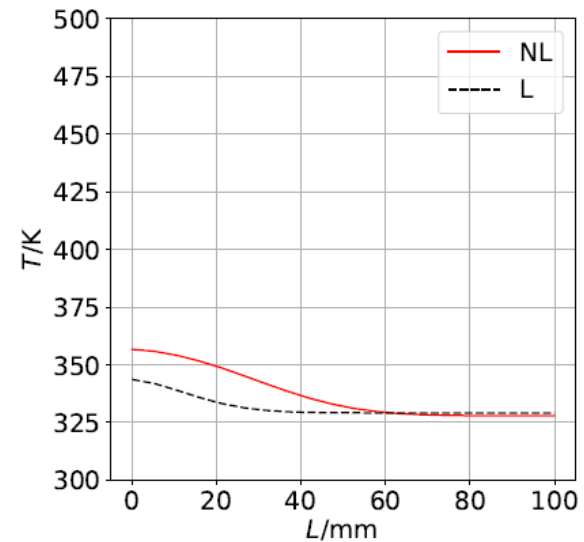
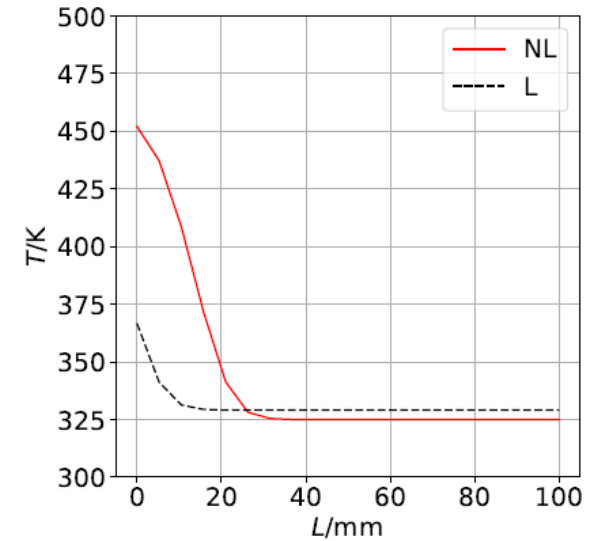
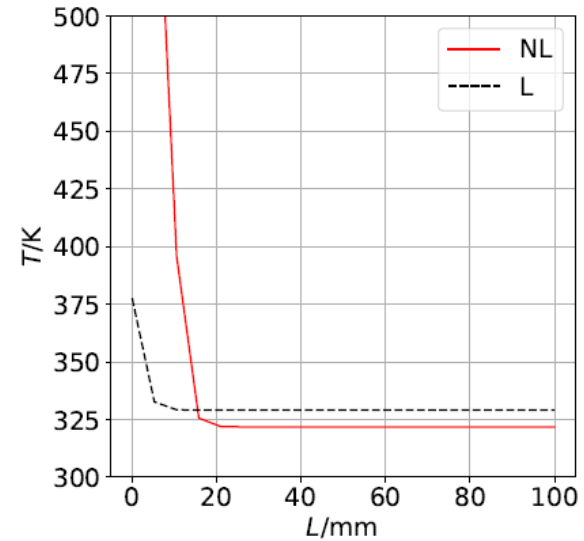
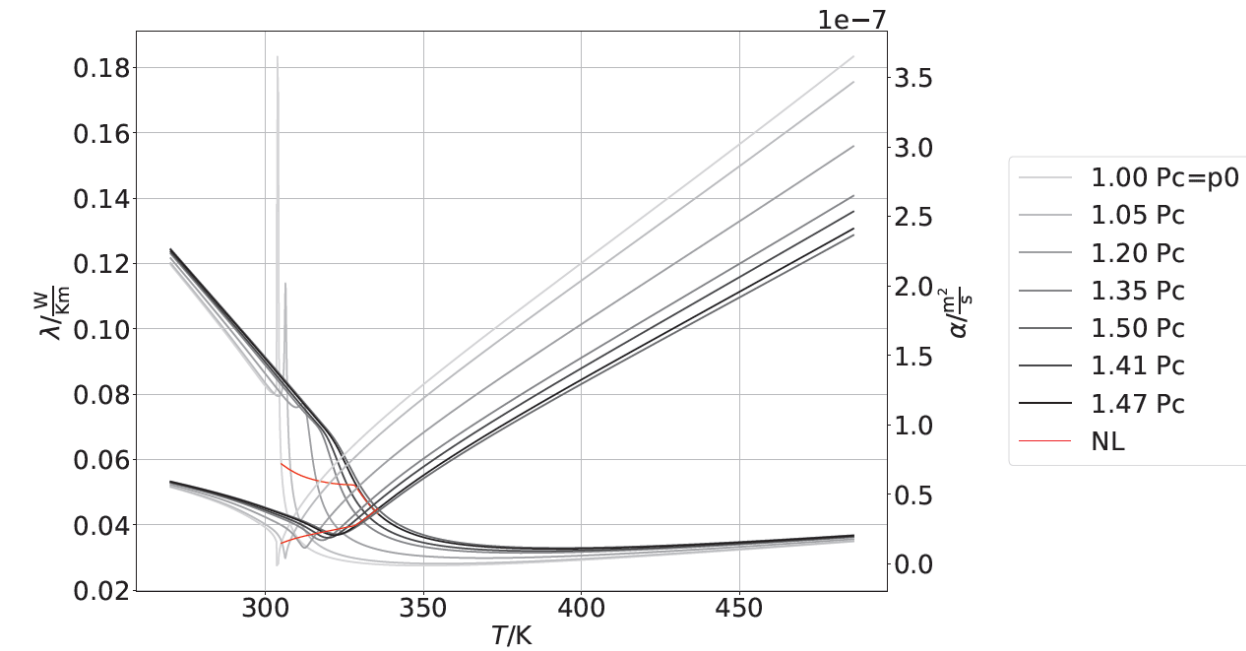
Nemlineáris eset vizsgálata



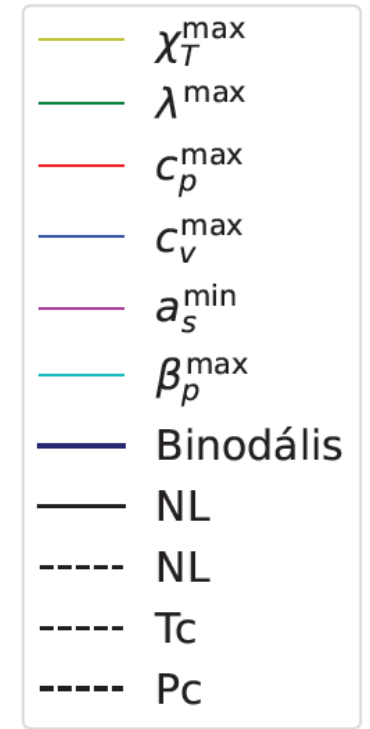
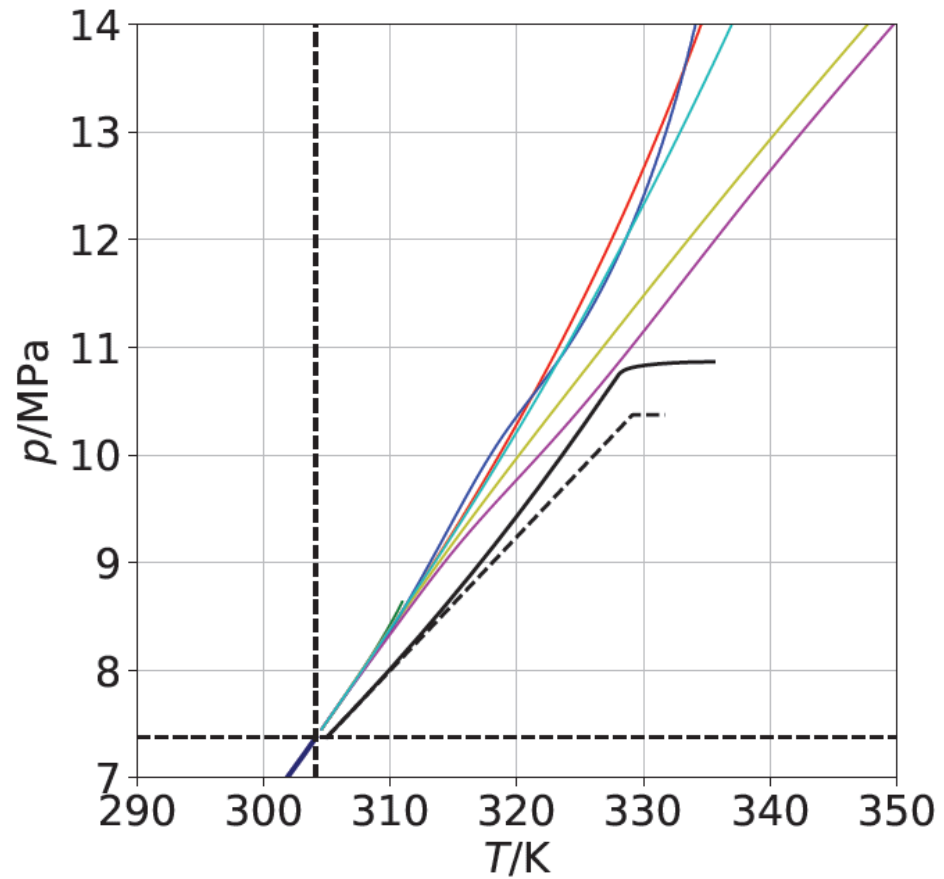
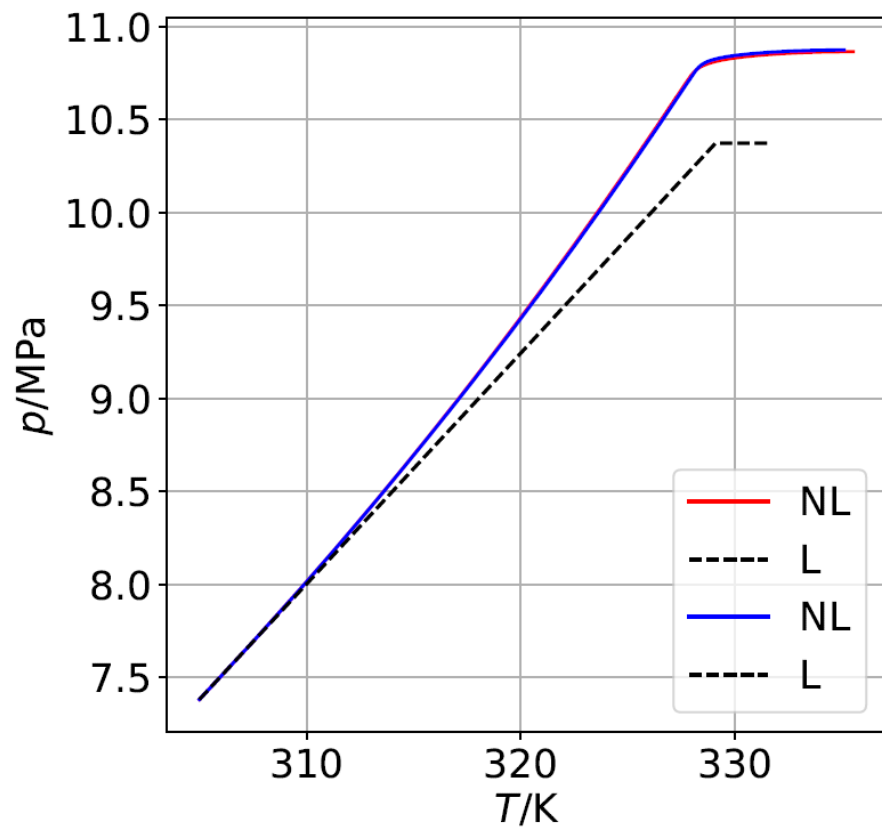
$$\begin{aligned}T^0 &= 305 \text{ K} \\p^0 &= 7.38 \text{ MPa} \\Q &= 38.5 \text{ J} \\t_h &= 10 \text{ s}\end{aligned}$$



Nemlineáris eset vizsgálata



Nemlineáris eset vizsgálata





Köszönöm a megtisztelő
figyelmet!
